

Mahanth Kumar Bhagyalakshmi

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Education

New York University (NYU) Tandon School of Engineering , M.S. in Cybersecurity	Sep 2026 – May 2028
• Coursework: AI in Cybersecurity, Computer Networking, Information security and privacy	
PES University , B.Tech in Computer Science and Engineering	Dec 2022 – May 2026
• Coursework: Data Structures, Design and Analysis of Algorithms, Computer Networks, Operating Systems, Database Management Systems, Web Technologies, Software Engineering, Augmented Reality (AR), Virtual Reality (VR)	

Skills

- **Programming:** Python, C, JavaScript, SQL
- **Cybersecurity & Networking:** Zero Trust, TCP/IP, OSINT, PowerShell
- **AI/ML:** PyTorch, TensorFlow, OpenCV, NLP, LLMs, RAG, Prompt Engineering
- **Web & Tools:** React JS, Three.js, Flask, Streamlit, Firebase, REST APIs, Git, Docker, MongoDB

Experience

SDE Intern , Neurofin AI Technologies Pvt. Ltd.	Nov 2025 – Jul 2026
Developed Generative AI solutions for financial institutions to automate transaction workflows and improve efficiency. Built AI-powered systems for financial analysis and automated compliance monitoring. Implemented intelligent document processing for extracting and structuring financial data.	
Summer Intern , CAVE Lab - PESU Center for IOT	June 2024 – July 2024
Contributed to the development of a web application utilizing React Three Fiber and Three.js to enable smooth handling of 3D models in both Augmented Reality (AR) and Virtual Reality (VR) environments.	
Summer Intern , MARS Lab - PESU Center for IOT	June 2025 – July 2025
Worked on a Comparative Study of Path Planning Algorithms for robotic sketching using the Niryo Ned 2 robotic arm. The focus was on evaluating and analyzing Path Planning Algorithm to improve robotic sketching performance.	

Projects

Automated Sketching Using Robotic Arm

- Designed and implemented a multimodal system to convert textual or audio descriptions into visual sketches using a robotic arm. Built a pipeline integrating NLP for input interpretation, text-to-image generation, outline extraction, and secure robotic arm control via serial communication. Optimized precision through custom path planning algorithms, resulting in high drawing accuracy and potential applications in forensics and creative automation.
- Tools Used: Python, OpenCV, PyTorch, Flask, Serial Communication, NLP, Speech Recognition, Custom Path Planning Algorithms

3D Viewer and Editor

Repo Link

- Developed an interactive web application enabling users to upload and manipulate GLTF/GLB files from local storage or Firebase. Implemented controls for light manipulation, scene properties (wireframe, autorotation, background, grid), mesh hierarchy, and mesh-specific editing (color, wireframe, transparency, position, rotation, scale). Enabled export of edited models back to storage and integrated augmented reality (AR) support for viewing models in AR. Additionally, provided functionality for selecting and playing animations within the scene.
- Tools Used: Three.js, Firebase, React, @react-three/fiber, @react-three/drei, @google/model-viewer

Publications

Automated Sketching with Robotic Arm: Integrating NLP for Visual Art Generation , 2026	Jul 2026
11th International Conference on Control and Robotics Engineering (ICCRE), IEEE	

Certifications

PwC Cyber Risk and Regulatory Launchpad Program (Trainee)	Jan 2025 – June 2025
Zscaler Zero Trust Associate (ZTCA) Certification , Zscaler Cyber Academy	July 2025 – July 2028
Introduction to Networking for Cyber Professionals , Zscaler Cyber Academy	July 2025 – July 2028
Fundamentals of Cybersecurity (EDU-102) , Zscaler Cyber Academy	July 2025 – July 2027

Extracurricular Activities

- Provided a practical demonstration and workshop on Virtual Reality (VR) applications for immersive learning and medical assessment, showcasing hands-on use of VR in education and diagnostics.
- Contributed to the coordination and successful management of the major HAL Aerothon '25 intercollegiate hackathon, assisting with event logistics and participant engagement.